



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
30-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS & TNAS]

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

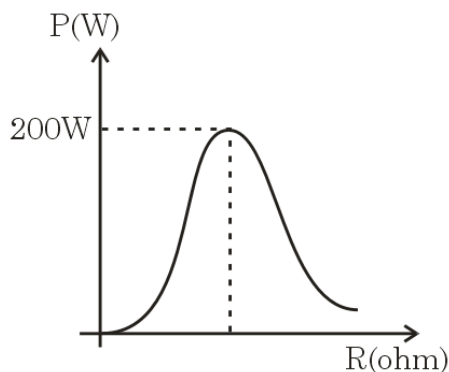
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

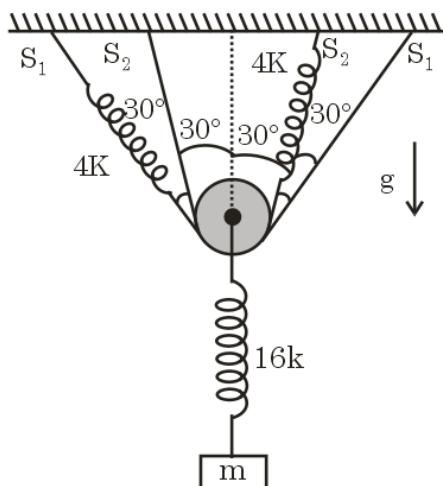
1. Moment of inertia of a uniform equilateral triangular lamina of mass 20 kg and side length 75 cm about an axis perpendicular to plane and passing through centroid (in kg-m^2) is :-
 (A) $\frac{360}{96}$ (B) $\frac{360}{32}$ (C) $\frac{360}{192}$ (D) $\frac{180}{192}$
2. A battery of emf 48 V unknown internal resistance is connected to a rheostat whose resistance can be varied in a long range. A graph is plotted between power developed across rheostat and resistance of rheostat which is shown in figure. The internal resistance of the battery in Ω is :-



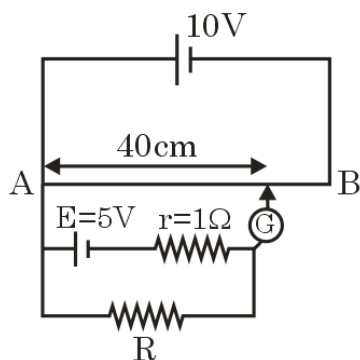
- (A) 1.44 (B) 2.88 (C) 5.76 (D) 11.52

Space for Rough Work

3. All the springs and pulleys are ideal and massless. If the block is slightly pulled vertically downward from its equilibrium position, then the time period of oscillation of the block is (assume strings are taught all the time)



- (A) $2\pi\sqrt{\frac{m}{4k}}$ (B) $\pi\sqrt{\frac{2m}{k}}$ (C) $\pi\sqrt{\frac{m}{2k}}$ (D) $2\pi\sqrt{\frac{m}{16k}}$
4. A potentiometer wire AB is 100 cm long and has total resistance of 10Ω . Find the value of unknown resistance R so that null point is obtained at a distance 40 cm from A.



- (A) 1Ω (B) 2Ω (C) 3Ω (D) 4Ω

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

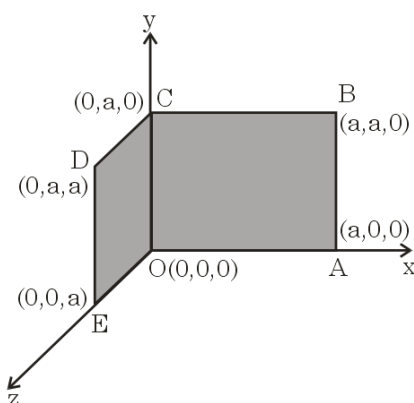
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

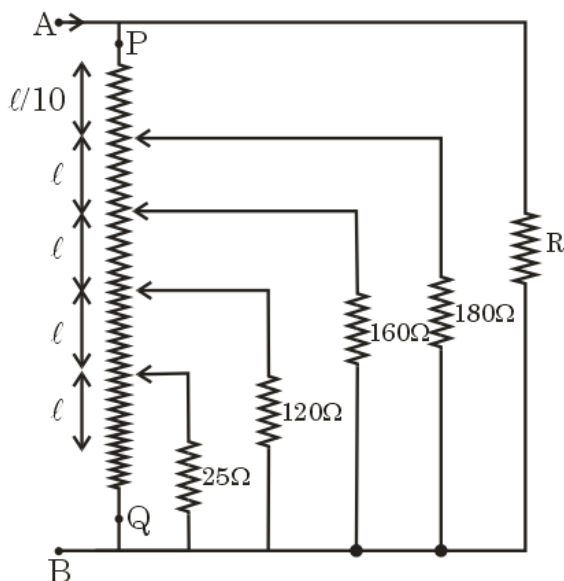
5. Two surface OABC and OCDE lies in the plane of xy and yz as shown in the figure. A charged particle 'q' lies in the space at a point P, if:



- (A) coordinates of 'P' is $(a - \Delta r, a - \Delta r, \Delta r)$ and $a \gg \Delta r$, then flux passing through surface OABC is $\frac{7q}{24\epsilon_0}$.
- (B) coordinates of 'P' is $(a - \Delta r, a - \Delta r, \Delta r)$ and $a \gg \Delta r$, then flux passing through surface OCDE is $\frac{q}{24\epsilon_0}$.
- (C) coordinates of 'P' is $(a + \Delta r, a + \Delta r, -\Delta r)$ and $a \gg \Delta r$, then flux passing through surface OABC is $\frac{q}{24\epsilon_0}$.
- (D) coordinates of 'P' is $(a + \Delta r, a + \Delta r, -\Delta r)$ and $a \gg \Delta r$, then flux passing through surface OCDE is $\frac{q}{24\epsilon_0}$.

Space for Rough Work

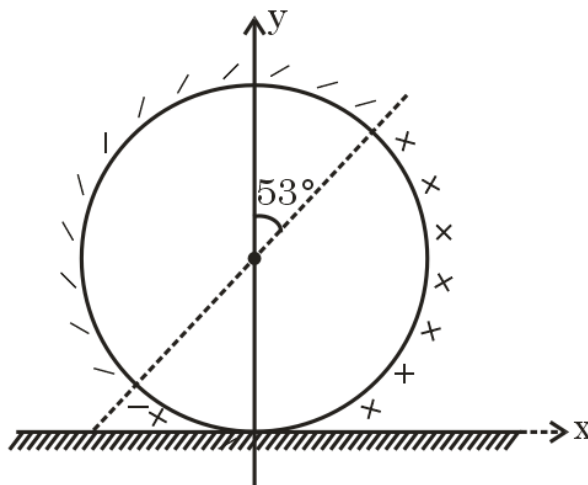
6. The resistance of wire PQ is $410\ \Omega$, which is tapped with the external wire as shown in the figure. The value of R is not known. If the potential difference between AB is 400 volts and the net current through the source battery is 6 ampere, then which of the following is/are correct :-



- (A) The resistance of unknown resistor R is $200\ \Omega$.
 (B) The equivalent resistance across AB is $\frac{200}{3}\ \Omega$.
 (C) The current in $25\ \Omega$ resistor is $\frac{2}{5}$ ampere.
 (D) The potential drop across $120\ \Omega$ resistor is 60 volts.

Space for Rough Work

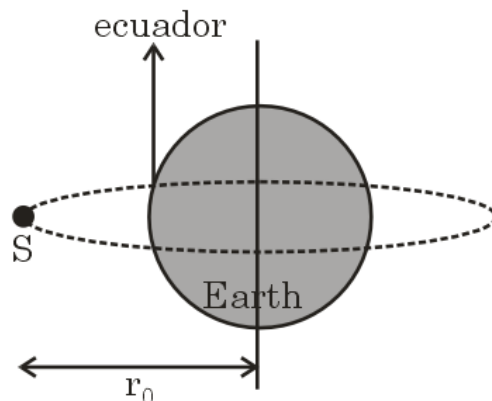
7. A non conducting uniform thin shell of mass m and radius R is having uniform charge density $+\sigma$ on one half and $-\sigma$ on another half as shown in figure. It is placed on a rough non conducting horizontal plane. At $t = 0$ a uniform electric field $\vec{E} = -E_0\hat{j}$ N/C is switched on and the solid sphere starts rolling without slipping. Then which of the following is/are correct.



- (A) The acceleration of the topmost point of the hollow sphere at $t = 0$, is $\frac{36}{25} \frac{\pi\sigma E_0 R^2}{m}$.
- (B) The magnitude of frictional force at $t = 0$ is $\frac{18}{25} \pi\sigma E_0 R^2$.
- (C) The acceleration of centre of mass of the hollow sphere at $t = 0$, is $\frac{12}{25} \frac{\pi\sigma E_0 R^2}{m}$.
- (D) The magnitude of frictional force at $t = 0$ is $\frac{9}{25} \pi\sigma E_0 R^2$.

Space for Rough Work

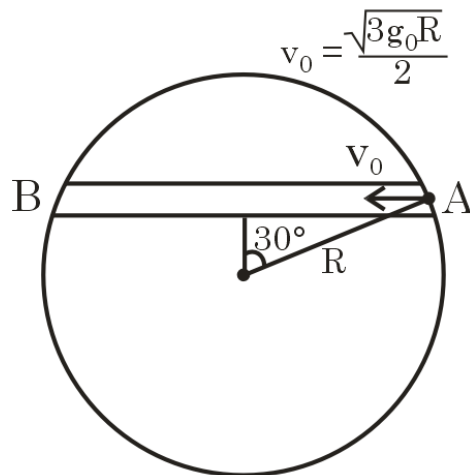
8. Consider a satellite 'S' which is just above the ECUADOR and rotating on equatorial plane of earth. The sense of rotation of earth and satellite are similar means both are rotating from west to east. The time period of satellite is 6 hours. At $t = 0$, satellite is just above the Brazil. Now choose the correct option (s).



- (A) The time after which the satellite will be above the ECUADOR again is 8 hours.
- (B) The above satellite may be a geostationary satellite.
- (C) The height of geostationary satellite above the sea level is 35,800 km approximately.
- (D) If the distance from the centre of earth to the satellite 'S' becomes $4r_0$, then the time after which the satellite will be above the ECUADOR again is 24 hours.

Space for Rough Work

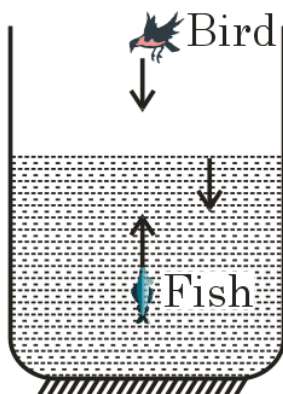
9. A tunnel is dug across the earth along a chord AB as shown, A particle is projected inside the tunnel from end 'A' with an initial velocity $v_0 = \frac{\sqrt{3g_0R}}{2}$, there is no friction between the tunnel and the particle. Then choose the correct option (s), (where g_0 = gravitational acceleration on the surface of Earth, R = radius of earth)



- (A) The time taken by the particle to move from end 'A' to other end 'B' of the tunnel is $\frac{\pi}{3} \sqrt{\frac{R}{g_0}}$.
- (B) The time taken by the particle to move from end 'A' to other end 'B' of the tunnel is $\frac{2\pi}{3} \sqrt{\frac{R}{g_0}}$.
- (C) The maximum speed of the particle inside the tunnel is $\sqrt{g_0R}$.
- (D) The maximum speed of the particle inside the tunnel is $\sqrt{2g_0R}$.

Space for Rough Work

10. A bird is flying downward with constant velocity 6 cm/s towards a fixed container filled with a liquid of refractive index $\frac{4}{3}$. A fish is moving upward with constant speed 4 cm/s towards the surface of liquid. Surface of liquid is also moving downward with constant velocity 2 cm/s shown in the figure. The bottom of container is silvered. Then :-



- (A) Speed of fish as seen by bird only after refraction is 9 cm/sec.
 (B) Speed of fish as seen by bird only after refraction is 8.5 cm/sec.
 (C) Speed of fish as seen by bird after reflection and then refraction is 3 cm/sec.
 (D) Speed of fish as seen by bird after reflection and then refraction is 2.5 cm/sec.

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

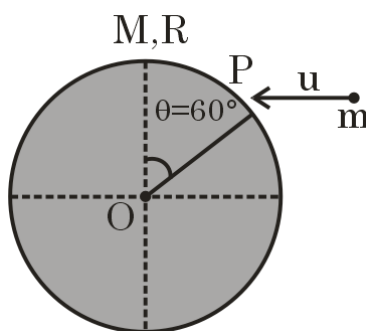
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A uniform circular disc of mass $M = 4\text{ kg}$ and radius $R = 25\text{ cm}$ is suspended in a vertical plane and hinged at its centre 'O'. It is free to rotate about a horizontal axis passing through its centre and perpendicular to its plane. A particle of mass $m = 0.5\text{ kg}$ moving horizontally with a velocity $u = 20\text{ m/s}$ collides with disc at point 'P' and gets stuck to it as shown. ($g = 10\text{ m/s}^2$)

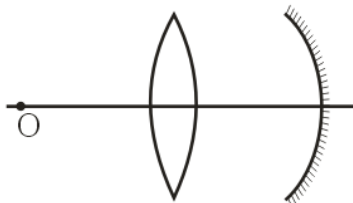


1. Find the angular velocity of the disc just after collision (in rad/sec)
2. The force on the disc due to hinge at its centre when the particle reaches the highest point (in newton).

Space for Rough Work

Paragraph for Questions 3 and 4

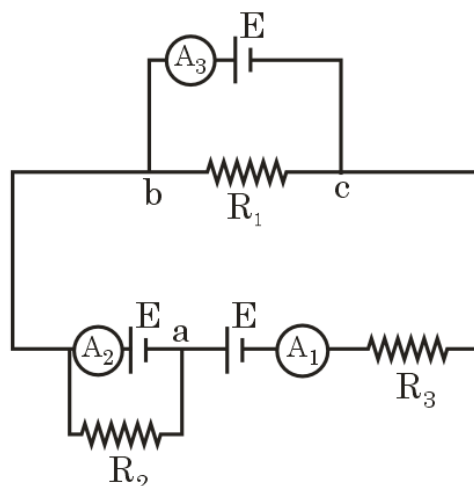
A converging lens of power $P_1 = +5D$ and a concave mirror of power $P_2 = +16D$ are placed at a separation 80 cm such that their principal axis coincide. A point object is placed left side of the lens on its principal axis at a distance X_1 from the lens such that image of the object coincides with the object itself. When the mirror is further shifted right side by a distance X_2 , the image of the object further coincides with the object.



3. The value of X_1 (in cm) is :
4. The value of X_2 (in cm) is:

Paragraph for Questions 5 and 6

In the circuit shown in figure, $R_1 = R_2 = R_3 = 3\Omega$ and e.m.f. of each cell is $E = 4V$ and negligible internal resistance. All ammeters are ideal.



5. Reading of ammeter A_1 in ampere is:
6. Potential difference (in Volt) between points a and b is

Space for Rough Work

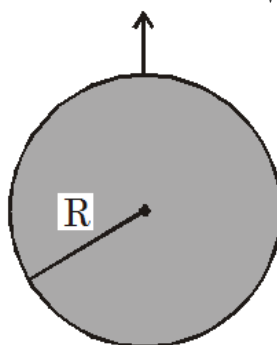
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

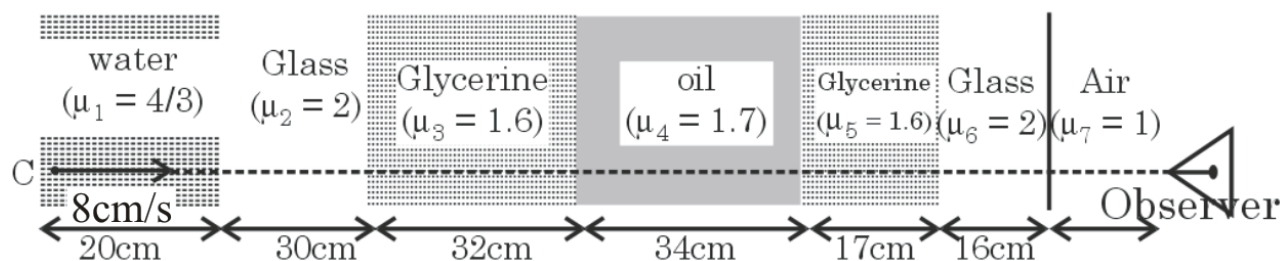
7. A small body of mass 'm' is projected from the surface of a given planet as shown in the figure. If small body has velocity which is just sufficient to get out from the gravity of planet then time taken by small body to reach at height of $3R$ from the surface of planet is $\frac{K}{3} \sqrt{\frac{2R^3}{GM}}$. Find the value of K.



Planet of mass M
and radius R

Space for Rough Work

8. A non-conducting solid sphere has a charge density given by $\rho = \rho_0 r^n$, where ρ_0 , n are constant and 'r' is the distance from the centre of sphere. If the magnitude of electric field at half of the radius is $\frac{1}{16}$ times that of the surfaces of nonconducting then find the value of n .
9. Find the velocity (in cm/s) of the car C as appeared to the observer shown in the figure. There are several parallel slabs of different medium placed between the car and the observer.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

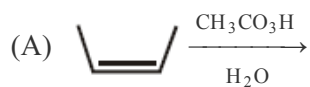
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
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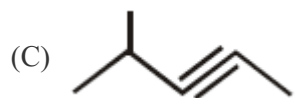
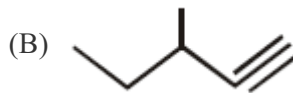
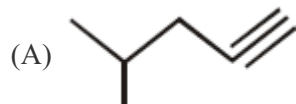
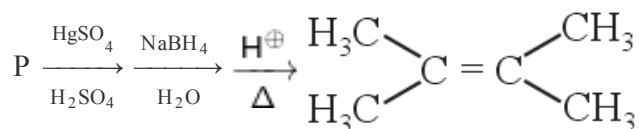
Negative Marks : -1 In all other cases

1. Which of the following will give non resolvable (meso) product :



Space for Rough Work

2. Which compound is suitable for 'P' in following reaction :



3. Select the incorrect statement :

- (A) MgFe_2O_4 can behave like a ferrimagnetic solid
 (B) The unit cell having crystal parameters, $a = b \neq c$, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$ is hexagonal
 (C) In ionic compounds having Frenkel defect, the ratio r^+/r^- is high
 (D) The co-ordination number of Na^+ ion in NaCl is 6

4. If FeO crystallizes rock salt structure in which O^{2-} ions occupy FCC lattice points. Due to some crystal defect, lattice becomes $\text{Fe}_{0.75}\text{O}$. Find number of effective Fe^{3+} ions per unit cell -

- (A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. The CORRECT order(s) regarding the atomic size is / are

(A) $Zr \simeq Hf$

(B) $Fe = Co = Ni$

(C) $Sc < Y < La$

(D) $Cu < Zn$

6. Which of the following compound(s) give(s) dibasic acid as a product on hydrolysis.

(A) P_4O_6

(B) SO_2Cl_2

(C) SF_6

(D) SeF_6

7. Which of the following ore(s) contain(s) copper metal?

(A) Dolomite

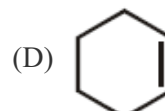
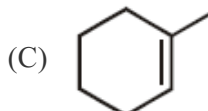
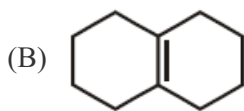
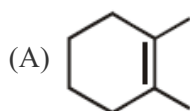
(B) Chalcopyrite

(C) Fool's gold

(D) Malachite

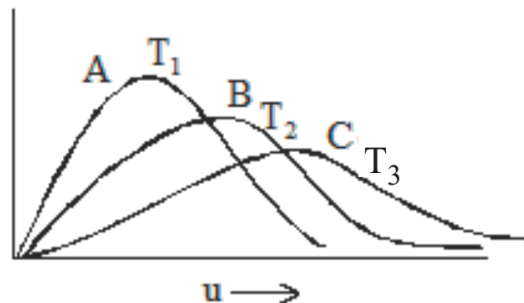
Space for Rough Work

8. Which of the following will give same product with H^+ , KMnO_4 , Δ and $\text{O}_3 / \text{Zn}, \text{H}_2\text{O}$?



9.

$$\frac{1}{N} \left(\frac{dN}{dU} \right)$$



Which of the following statement is correct.

- (i) If A & B are same gases then $T_2 > T_1$
(ii) If $T_1 = T_2$ then $A_{(g)}$ may be $SO_{2(g)}$ & $B_{(g)}$ may be $CH_{4(g)}$
(iii) If B & C are same gases then $T_3 > T_2$
(iv) If $T_1 = T_2 = T_3$ then $A_{(g)}$ may be CH_4 , $B_{(g)}$ may be SO_2 & $C_{(g)}$ may be $SO_{3(g)}$

- (A) i, ii & iii only
(B) i, ii, & iv only
(C) All are correct
(D) ii & iii only

10. Two volatile liquids A & B are mixed to form an ideal solution. If $P_A^0 = 100$ mm of Hg & $P_B^0 = 150$ mm of Hg then identify the correct option(s) if 10 moles of A are mixed with 40 moles of B such that moles vaporised are very small.

- (A) Total vapour pressure will be 140 mm of Hg.
- (B) Mole fraction of A in vapour state will be less than 0.5.
- (C) Mole fraction of B in vapour state is equal to $\frac{6}{7}$
- (D) Mole % of B in vapour state will be greater than 80%.

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
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Paragraph for Questions 1 and 2

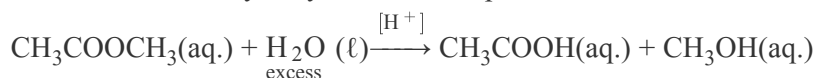
The complex compounds show space isomerism when the ligand species have different spatial arrangement surrounding the CMA

1. What is the number of stereo isomeric forms for the complex $[\text{Pt}(\text{py})(\text{NH}_3)(\text{Br})(\text{I})]$
2. How many stereo isomers are possible for $[\text{Cr}(\text{H}_2\text{O})_2(\text{C}_2\text{O}_4)_2]^-$?

Space for Rough Work

Paragraph for Questions 3 and 4

Kinetics of acidic hydrolysis of ester is pseudo I order reaction



Rate law is given by : $r = K[\text{H}^+][\text{H}_2\text{O}][\text{CH}_3\text{COOCH}_3]$; $K = 1.8 \times 10^{-3} \text{ M}^{-2} \text{ sec}^{-1}$

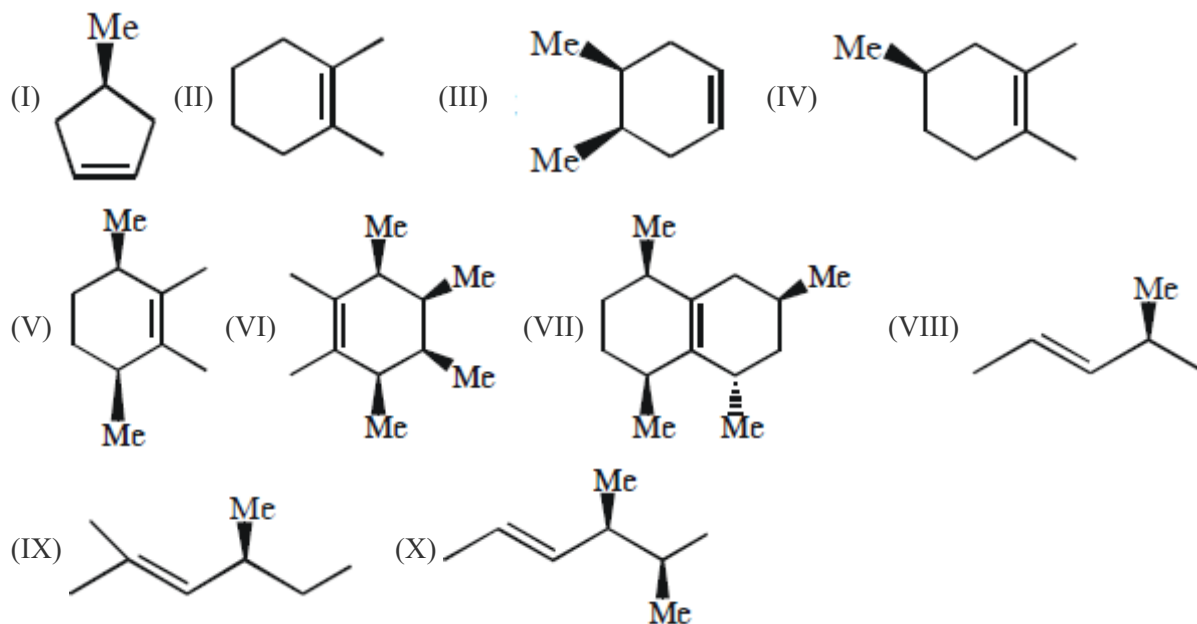
where H^+ ion concentration is given by acid catalyst. In an experiment 1M $\text{CH}_3\text{COOCH}_3$ is hydrolysed using 0.1M HCl as catalyst. [Use molarity of $\text{H}_2\text{O} = \frac{1000}{18} \text{ M}$]

3. Reaction mixture is titrated against standard NaOH solution at different times. If initially 10 ml of NaOH was required while after long time 100 ml of NaOH is required then find volume of NaOH (in ml) required at 230.3 sec.
4. Calculate the time at which ester concentration reduce to 0.25M (in seconds) [$\ln 2 = 0.7$]

Space for Rough Work

Paragraph for Questions 5 and 6

Among the following alkenes :



5. How many alkenes which give meso compound on ozonolysis in presence of Zn?
6. From the following options identify alkenes which give optically active compound as one of the product on ozonolysis in presence of Zn.

Space for Rough Work

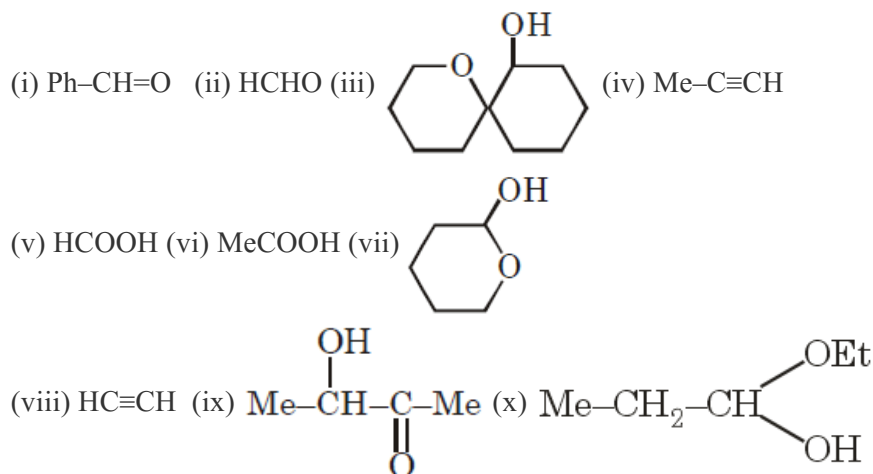
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Total number of complex compound which have low spin as well as Diamagnetic.
 $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{NiF}_6]^{2-}$, $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{Ni}(\text{en})_3]^{3+}$, $[\text{PtCl}_4]^{2-}$
8. Total number of species in which highest occupied molecular orbital has gerade type symmetry.
 B_2 , N_2 , C_2 , N_2^{2+} , O_2 , O_2^{2-} , N_2^{2-}
9. How many of the following will give silver mirror with Tollen's reagent ?



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. $\int \left(2e^{\tan^{-1}x + x^2 + \ln x} + \frac{(2x^3 + 2x + 1)}{\left(x + \frac{1}{x}\right)} e^{\tan^{-1}x + x^2 + \ln x} \right) dx$ is equal to

(where C is constant of integration)

(A) $\frac{x^3}{3} e^{\tan^{-1}x + x^2 + \ln x} + C$

(B) $\frac{x^2}{2} e^{\tan^{-1}x + x^2 + \ln x} + C$

(C) $xe^{\tan^{-1}x + x^3 + \ln x} + C$

(D) $x^2 e^{\tan^{-1}x + x^2} + C$

Space for Rough Work

2. The function $f(x) = \alpha x^2 - \beta x - 4x^3 + \gamma$, where $\alpha, \beta, \gamma \in \mathbb{R}$, has local minima at $P(\log_2 a, f(\log_2 a))$ & local maxima at $Q(\log_2 a^2, f(\log_2 a^2))$. If the graph of $f(x)$ changes concavity about the point $R\left(\frac{3}{4}, f\left(\frac{3}{4}\right)\right)$, then which of the following conic section can have eccentricity 'a' -
- (A) circle (B) parabola
(C) ellipse (D) hyperbola
3. If m is the slope of a line which is tangent to $y^3 = x^4$ & a normal to $x^2 - 2x + y^2 = 0$, then $\left(\frac{3m}{4}\right)^3$ is equal to ($m \neq 0$)
- (A) 3 (B) $\frac{4}{3}$ (C) 4 (D) $\frac{3}{4}$
4. For the equation $b\sqrt{1-a^2} - a\sqrt{1-b^2} = \sin\left(\frac{b-a}{\sqrt{1-x^2}}\right)$, where $0 < a < b < 1/2$, then which of the following is true -
- (A) there is no real root of the equation in (a, b)
(B) there is exactly one real root of the equation in (a, b)
(C) there is atleast one root of the equation in (a, b)
(D) none of these

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Consider twice derivable bijective function defined on $[0,1]$ such that $f(\alpha) = \frac{1}{3}$ and $f(f(\beta)) = \frac{1}{3}$. If $f''(x) > 0 \forall x \in \left[0, \frac{1}{2}\right)$ and $f''(x) < 0 \forall x \in \left(\frac{1}{2}, 1\right]$ with $f\left(\frac{1}{2}\right) = \frac{1}{2}$. Which the following holds good ?
- (A) $\beta > \alpha$ implies $f(x)$ is an increasing function
- (B) $\beta > \alpha$ implies $f(x)$ is a decreasing function
- (C) $2f\left(\frac{1}{4}\right) < f(0) + f\left(\frac{1}{2}\right)$
- (D) $2f\left(\frac{3}{4}\right) < f\left(\frac{1}{2}\right) + f(1)$

Space for Rough Work

6. For the function $f(x) = e^{e^x} - \tan^{-1}x - 3$ which of the following holds good ?
- (A) $f(x) = 0$ has exactly one positive real root
- (B) $y = f(x)$ has exactly one minima at negative x
- (C) $y = f(x)$ has exactly one point of minima at positive x
- (D) $y = f(x)$ is strictly increasing function
7. Let x_1, x_2, x_3 be the roots of the equation $x(3 - x^2) = 1$, then
- (A) exactly one of x_1, x_2, x_3 is real
- (B) exactly one of x_1, x_2, x_3 is rational
- (C) the value of $\sum_{i=1}^3 \left(\lim_{m \rightarrow \infty} \left(\lim_{n \rightarrow \infty} \cos(\pi n! x_i) \right)^m \right)$ where $m, n \in \mathbb{N}$, is 3
- (D) the value of $\sum_{i=1}^3 \left(\lim_{m \rightarrow \infty} \left(\lim_{n \rightarrow \infty} \cos(\pi n! x_i) \right)^m \right)$ where $m, n \in \mathbb{N}$, is 0
8. Let $f(x)$ be a non-negative continuous and bounded function for all $x \geq 0$.
If $(\cos x) f'(x) \leq (\sin x - \cos x) f(x) ; \forall x \geq 0$, then which of the following is / are correct ?
- (A) $f(6) + f(5) > 0$
- (B) $x^2 - 3x + 2 + f(7) = 0$ has 2 distinct solutions
- (C) $f(5)f(7) - f(6)f(5) = 0$
- (D) $\lim_{x \rightarrow 6} \frac{f(x) - \sin(\pi x)}{(x - 6)} = 1$

Space for Rough Work

9. Let $g(x)$ be a polynomial of degree 2018 and $f(x) = \begin{cases} g\left(\frac{1}{x^5}\right)e^{-\frac{1}{x^6}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ then

(A) $f(x)$ is continuous at $x = 0$

(B) $f(x)$ is non-differentiable at $x = 0$

(C) $f'(0) = \lim_{x \rightarrow 0} f(x)$

(D) $\lim_{x \rightarrow 0} f(x)$ does not exist

10. Let $f(x) = \begin{cases} -2 & \text{for } -3 \leq x \leq 0 \\ x-2 & \text{for } 0 < x \leq 3 \end{cases}$, $g(x) = [f(|x|)] + [f(x)]$ where $[.]$ denotes the greatest integer function. Which of the following statement(s) is/are true ?

(A) $g(1.7) = -1$

(B) $g(x)$ is not differentiable at any point

(C) $g(x)$ is continuous at all non integral points

(D) $g(x)$ is differentiable at some points

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A Rod of length 10cm slides between two lines, x-axis & $y = -\sqrt{3}x$. If locus of mid point of rod is curve 'S'.

1. x-intercept of S is
2. If maximum distance between origin and curve S is equal to k then value of $k\sqrt{3}$ is

Space for Rough Work

Paragraph for Questions 3 and 4

Let $f(x) = \sin^{-1}(\sin^x)$, $g(x) = \sin^{-1}\left(\frac{3^x + 3^{-x}}{4}\right)$ and $h(x) = f(x) - g(x)$

3. If range of $g(x)$ is $\left[\frac{\pi}{a}, \frac{\pi}{b}\right]$, then value of $(a + b)$ is
4. Number of solution(s) of $h(x) = 0$ is

Paragraph for Questions 5 and 6

Let $f : [0, \infty) \rightarrow [2, \infty)$ be a differentiable increasing and onto function satisfying

$$f^2(x) - 2f(x) - \sqrt{x} = f^2(y) - 2f(y) - \sqrt{y}, \forall x, y \in [0, \infty)$$

5. The value of $\lim_{x \rightarrow 0^+} \frac{f(x) - 2}{\sqrt{x}}$ is equal to -
6. If $g(x)$ be the inverse of $f(x)$, then $g'(4)$ is equal to -

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. A function f satisfies $f(3x) = 3f(x) \forall x \in [1, \infty)$ and $f(x) = 1 - |x - 2|$ for $x \in [1, 3]$. The smallest x for which $f(x) = f(2001)$ is
 8. For $\ell, m, n \in \mathbb{R}$, p and q are the value of ℓ and m respectively such that equation $\ell(13 + 11n) + m(19 + 17n) - 4(8 + 7n) = 0$ is satisfied $\forall n \in \mathbb{R}$ then $(p + q)$ equals
 9. The value of k ($k > 0$) such that the length of the longest interval in which the function $f(x) = \sin^{-1} |\sin kx| + \cos^{-1} (\cos kx)$ is constant is $\frac{\pi}{4}$ will be
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Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work